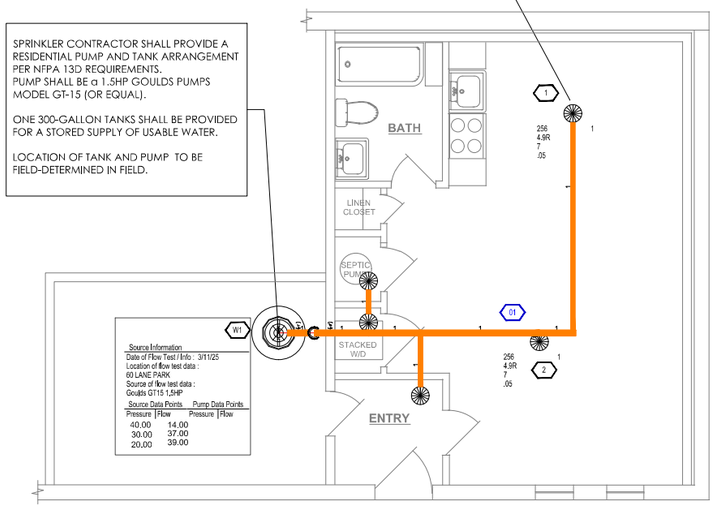
COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO KAHAN
FIRE PROTECTION
NO. 48398
REGISTERED PROFESSIONAL ENGINEER
04/11/2025

PROVIDE RESIDENTIAL PENDENT SPRINKLERS (K=4.9) WHERE INDICATED THROUGHOUT. MAXIMUM SPACING USED IN THIS DESIGN IS 16'x16' (REQUIRING 7 PSI MIN)

SPRINKLER CONTRACTOR SHALL PROVIDE A RESIDENTIAL PUMP AND TANK ARRANGEMENT PER NFPA 13D REQUIREMENTS. PUMP SHALL BE a 1.5HP GOULDS PUMPS MODEL GT-15 (OR EQUAL).

ONE 300-GALLON TANKS SHALL BE PROVIDED FOR A STORED SUPPLY OF USABLE WATER.

LOCATION OF TANK AND PUMP TO BE FIELD-DETERMINED IN FIELD.



Source Information
Date of Flow Test / Info : 3/11/25
Location of flow test data :
60 LANE PARK
Source of flow test data :
Goulds GT15 1.5HP

Source Data Points		Pump Data Points	
Pressure	Flow	Pressure	Flow
40.00		14.00	
30.00		37.00	
20.00		39.00	

PARTIAL BASEMENT SPRINKLER LAYOUT
SCALE: 1/4" = 1'-0"

SPRINKLER PIPE LEGEND

NEW STEEL PIPING = (SCH 10/40 STEEL W/ROLL GRVD FITTINGS)

NEW CPVC PIPING = (CPVC PIPING W/CPVC FITTINGS)

EXISTING PIPING =

NEW TAP =

Design Area 1 — Wet System
ADU BASEMENT
Demand Calculations using Hazen-Williams Method
Occupancy Classification: LH
Design Area Density: .05
Additional Outside Hose:
Design Area Size:
Notes: 2 HEAD

Job: 60 LANE PARK
Address: 60 LANE PARK
City: BRIGHTON State: MA Zip/Postal Code: 02135
Contract #:
Contractor: CAP DESIGN GROUP Date: 04/09/2025 Designer: JK
Address: 34 WALKERS BROOK DR. #1036
City: READING State: MA Zip/Postal Code: 02126
Phone: (617) 644-0014 Fax: Email:
Approving Authority: Boston

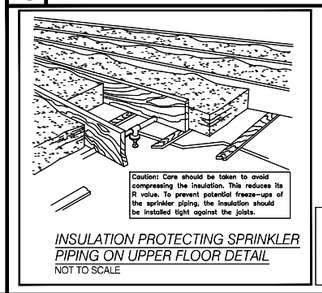
Design Defaults — Standards: NFPA13D

Default Sprig Size:	1	Default Drop Size:	1
Default Sprig Material:	CPVC	Default Drop Material:	CPVC
Default Sprig Elevation:	0	Default Drop Elevation:	0

Calculation results for Design Area 1 - ADU BASEMENT
This system as shown on CAP DESIGN GROUP company print no dated 04/09/2025
for 60 LANE PARK at 60 LANE PARK
contract no is designed to discharge at a rate of 0.05 gpm/ft² (L/min/m²) of floor area over a maximum area of 512 ft² when supplied with water at a rate of 26.3 gpm at 15 psi at the base of the riser.
Hose stream allowance of is included in the above.
Occupancy classification: LH
Commodity classification:
Maximum storage height:
Storage arrangement:
Number of heads flowing: 2
System Type: Wet
Maximum velocity: 8.87 ft/s

Flow from In-Rack sprinklers:	0 gpm	Pressure Required at Source:	15 psi
Flow from Overhead sprinklers:	26.3 gpm	Pressure Available at Source:	35.6 psi
Flow from Inside Hoses:	0 gpm	Surplus Pressure at Source:	20.6 psi
Flow from Outside Hoses:	0 gpm		
Other fixed flows:	0 gpm		
Total flow in system piping:	26.3 gpm		
Additional flow at/beyond source:	0 gpm		
Total of all flows:	26.3 gpm		

5 FREEZE PROTECTION

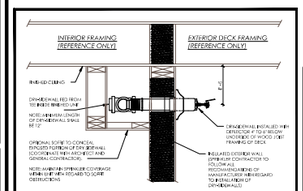


INSULATION PROTECTING SPRINKLER PIPING ON UPPER FLOOR DETAIL
NOT TO SCALE

NOTE: ALL COMBUSTIBLE CONCEALED SPACES, INCLUDING INTERSTITIAL SPACES, VOIDS UNDER STAIRS, ETC. WHICH ARE GREATER THAN 6" IN DEPTH, SHALL BE COMPLETELY FILLED WITH NON-COMBUSTIBLE INSULATION BY THE GENERAL CONTRACTOR OR ARE REQUIRED TO BE PROVIDED WITH SPRINKLER PROTECTION IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D (2013 EDITION). ANY SUCH AREA CREATED DURING CONSTRUCTION SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER IN WRITING BY THE GENERAL CONTRACTOR PRIOR TO INSTALLATION OF GYPSUM BOARD WALLS AND/OR CEILINGS.

NOTE: IT IS THE BUILDING OWNER'S RESPONSIBILITY TO PROVIDE SUFFICIENT HEAT WITHIN ALL AREA CONTAINING SPRINKLER PIPING TO PREVENT FREEZE-UPS. CAP DESIGN GROUP, INC. TAKES NO RESPONSIBILITY FOR FREEZE-UPS ON THIS SPRINKLER SYSTEM. ANY AREAS OF CONCERN SHALL BE BROUGHT TO THE ENGINEER'S ATTENTION IN WRITING PRIOR TO INSTALLATION.

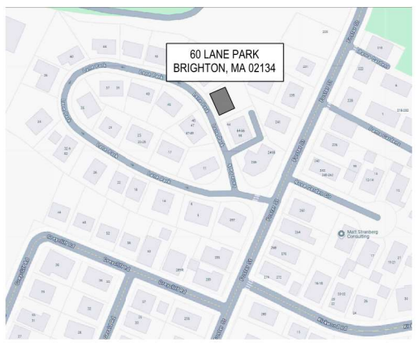
6 DRY SIDEWALL DETAIL



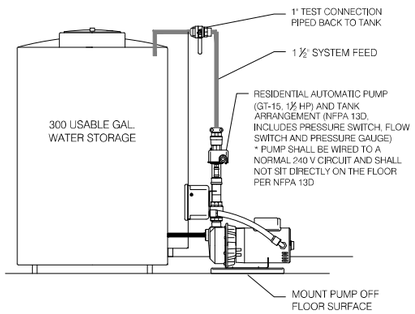
TYPICAL DRY-SIDEWALL INSTALLATION FOR EXTERIOR OVERHANGS
NOT TO SCALE

NOTE: AT THE TIME OF THIS SPRINKLER SYSTEM DESIGN, THE RCP AND LIGHTING SELECTION HAD NOT BEEN COMPLETED. SPRINKLER HEADS HAVE BEEN POSITIONED IN ACCORDANCE WITH CODE REQUIREMENTS AND HAVE NOT BEEN COORDINATED WITH LIGHTING OR OTHER SURFACE-MOUNTED FIXTURES. ANY LIGHTS OR OTHER FIXTURES THAT CAUSE OBSTRUCTIONS TO SPRINKLER HEAD SPRAY PATTERNS WILL REQUIRE ADDITIONAL SPRINKLER HEADS AT AN ADDITIONAL COST TO THE BUILDING OWNER.

1 PROJECT INFORMATION



2 PUMP AND TANK DETAIL



3 GENERAL DETAILS

TABLE 8.6.5.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-6"	0'-2 1/2"
1'-6" TO LESS THAN 2'-0"	0'-5 1/2"
2'-0" TO LESS THAN 2'-6"	0'-8 1/2"
2'-6" TO LESS THAN 3'-0"	0'-11 1/2"
3'-0" TO LESS THAN 3'-6"	0'-9 1/2"
3'-6" TO LESS THAN 4'-0"	1'-0"
4'-0" TO LESS THAN 4'-6"	1'-2"
4'-6" TO LESS THAN 5'-0"	1'-4 1/2"
5'-0" TO LESS THAN 5'-6"	1'-6"
5'-6" TO LESS THAN 6'-0"	1'-8"
6'-0" TO LESS THAN 6'-6"	2'-0"
6'-6" TO LESS THAN 7'-0"	2'-4"
7'-0" TO LESS THAN 7'-6"	2'-11"

NFPA 13R TABLE 8.6.5.1.2 + FIGURE 8.6.5.1.2.
(B) STANDARD PENDENT AND UPRIGHT SPRAY SPRINKLERS

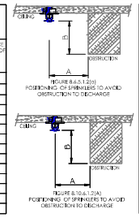


FIGURE 8.6.5.1.2(b)
POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

FIGURE 8.10.6.1.2(A)
POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

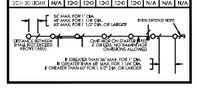
TABLE 8.10.6.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

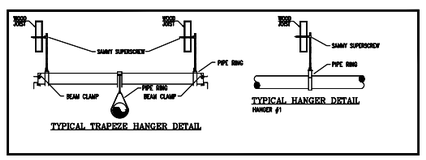
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-6"	0'-4"
1'-6" TO LESS THAN 2'-0"	0'-11"
2'-0" TO LESS THAN 2'-6"	0'-11"
2'-6" TO LESS THAN 3'-0"	0'-11"
3'-0" TO LESS THAN 3'-6"	0'-3"
3'-6" TO LESS THAN 4'-0"	0'-5"
4'-0" TO LESS THAN 4'-6"	0'-8"
4'-6" TO LESS THAN 5'-0"	0'-7"
5'-0" TO LESS THAN 5'-6"	0'-7"
5'-6" TO LESS THAN 6'-0"	0'-7"
6'-0" TO LESS THAN 6'-6"	0'-9"
6'-6" TO LESS THAN 7'-0"	0'-11"
7'-0" AND GREATER	1'-0"

NFPA 13R TABLE 8.10.6.1.2 + FIGURE 8.10.6.1.2
(A) RESIDENTIAL PENDENT AND UPRIGHT SPRAY SPRINKLERS

MAX. HANGER SPACING

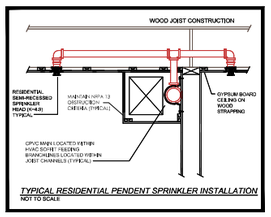
PIPE TYPE	NOMINAL PIPE SIZE (IN)								
	1	1	1/4	1 1/2	2	2 1/2	4	6	8
SCH 7,10,40	N/A	12-0	12-0	12-0	12-0	12-0	12-0	12-0	12-0
SCH 30 LIGHT	N/A	12-0	12-0	12-0	12-0	12-0	N/A	N/A	N/A





TYPICAL TRAPEZE HANGER DETAIL

TYPICAL HANGER DETAIL
HANGER #1



TYPICAL RESIDENTIAL PENDENT SPRINKLER INSTALLATION
NOT TO SCALE

4 GENERAL NOTES

FIRE PROTECTION NOTES:

THE PURPOSE OF THIS FIRE PROTECTION DRAWING AND THE ASSOCIATED FIRE PROTECTION DESIGN NARRATIVE IS TO INDICATE THE PROPOSED RESIDENTIAL SPRINKLER SYSTEM TO BE INSTALLED WITHIN THE RENOVATED MULTI-UNIT BUILDING LOCATED AT **60 LANE PARK BRIGHTON MA 02135.**

THIS BUILDING IS COMPRISED OF A BASEMENT LEVEL WITH TWO RESIDENTIAL FLOORS ABOVE. THE BUILDING IS LESS THAN 50-FT WITH FIRE DEPARTMENT VEHICULAR ACCESS AND THEREFORE DOES NOT REQUIRE A STANDPIPE SYSTEM.

THIS RESIDENTIAL SPRINKLER SYSTEM HAS BEEN DESIGNED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D (2013 EDITION) FOR A RESIDENTIAL BUILDING UP TO 4 STORIES (HIGHEST OCCUPIED FLOOR NO HIGHER THAN 60 FEET) AND NOT EXCEEDING 12,000 GROSS SQFT PER INDIVIDUAL UNIT.

THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13D (2013 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR, 2015 EDITION) AND **BOSTON** FIRE DEPARTMENT REQUIREMENTS.

THIS SYSTEM WILL BE SUPPLIED BY A RESIDENTIAL PUMP AND TANK SYSTEM WHICH HAS BEEN SIZED TO ACCOMMODATE THE FLOW FROM TWO RESIDENTIAL SIDEWALL SPRINKLER HEADS FOR A DURATION OF 10-MINTUTES IN ACCORDANCE WITH THE REQUIREMENTS OF THE MASSACHUSETTS ADOPTIONS TO THE MASSACHUSETTS STATE BUILDING CODE. THE STORED CAPACITY OF THE WATER TANKS IS 300 MINIMUM USABLE GALLONS AND THE PUMP SELECTED IS A GOULDS PUMPS MODEL GT15, 1.5 HP PUMP.

THE SYSTEM HAS BEEN HYDRAULICALLY CALCULATED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D, INCLUDING THE 2(TWO) HYDRAULICALLY MOST DEMANDING HEADS IN A SINGLE COMPARTMENT BASED ON THE REQUIREMENTS OF THE SPECIFIC SPRINKLER HEAD AND THE SPACING USED IN THIS DESIGN (16'x16' REQUIRING A MINIMUM OF 13 GPM @ 7.0 PSI). REMOTE AREAS, DENSITIES AND HOSE STREAM ALLOWANCES ARE INDICATED ON THIS DRAWING ALONG WITH SYSTEM DEMANDS AT CONNECTION TO STREET. CALCULATION RESULTS HAVE BEEN COMPARED TO RECENT HYDRANT FLOW TEST INFORMATION OBTAINED FROM THE **BOSTON** WATER & SEWER DEPARTMENT. CALCULATIONS HAVE BEEN SUBMITTED WITH THIS DRAWING TO THE **BOSTON** FIRE DEPARTMENT FOR REVIEW.

INSTALLATION NOTES:

ALL WORK SHALL BE PERFORMED BY A MASSACHUSETTS LICENSED SPRINKLER CONTRACTOR. THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13D (2013 EDITION), MASSACHUSETTS STATE BUILDING CODE (2015 EDITION, CHAPTER 9) AND THE **BOSTON** FIRE DEPARTMENT.

THE ARCHITECTURAL BACKGROUND OF BUILDING MAY DIFFER SLIGHTLY FROM ACTUAL LAYOUT. DRAWINGS ARE NOT INTENDED TO SHOW ALL OFFSETS AND PIPING ELEVATION CHANGES. CONTRACTOR SHALL FIELD VERIFY ALL MEASUREMENTS PRIOR TO FABRICATION.

CONTRACTOR SHALL HYDROSTATICALLY TEST ALL SPRINKLER PIPING AT 200 PSI FOR 2 HOURS AND IS RESPONSIBLE FOR THE COMPLETION OF ALL ABOVE GROUND TEST CERTIFICATES. SUPPLIED TO THE OWNER.

ALL PIPING INSTALLED THROUGHOUT THE RESIDENTIAL AREAS OF THE BUILDING SHALL BE UL LISTED CPVC SPRINKLER PIPING. ALL PIPING WITHIN THE STORAGE AND UTILITY AREAS OF THE BASEMENT LEVEL WILL BE BLACK STEEL PIPING. ALL PIPING SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D(2013 EDITION) AND ALL MANUFACTURERS INSTALLATION RECOMMENDATIONS. ALL PIPING SHALL BE PITCHED TO DRAIN WITH LOW-POINT DRAINS AT SECTIONS OF PIPING SUBJECT TO WATER TRAPPING. SPRINKLER CONTRACTOR SHALL PROVIDE PROTECTIVE PLATES WHERE CPVC PIPING IS RUN THROUGH STUDS TO PREVENT PUNCTURING OF THE SPRINKLER PIPING DURING DRYWALL INSTALLATION AS REQUIRED BY NFPA STANDARDS.

ALL SPRINKLER HEADS WITHIN RESIDENTIAL AREAS OF THE BUILDING SHALL BE RESIDENTIAL PENDENT AND SIDEWALL SPRINKLERS. SPRINKLER HEADS WITHIN THE BASEMENT UTILITY AREAS SHALL BE QUICK-RESPONSE COMMERCIAL UPRIGHT SPRINKLERS. ALL HEADS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND THE REQUIREMENTS OF NFPA 13D (2013 EDITION).

BUILDING OWNER IS RESPONSIBLE FOR MAINTAINING THIS SPRINKLER SYSTEM IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 25, INCLUDING THE PROVISION OF HEAT IN ALL AREAS CONTAINING SPRINKLER PIPING AND HEADS TO PREVENT PIPE FROM FREEZING. CAP DESIGN GROUP AND THE ENGINEER OF RECORD TAKE NO RESPONSIBILITY FOR ANY DAMAGES CAUSED BY FREEZE UPS.

SPRINKLER SYMBOLS

SYM	CNT	POSITION	FINISH	TEMP	K	NPT	MFG.
		PEND	WHITE	155	5.60	1/2"	Victaulic or Equivalent
	5	RES. PEND	WHITE	155	4.90	1/2"	Victaulic or Equivalent
	4	SIDE	CHROME	155	5.60	1"	Victaulic or Equivalent

REVISIONS

DATE	DESCRIPTION	BY

CAP DESIGN GROUP:

CAP DESIGN GROUP
www.capcofire.com
34 WALKERS BROOK DR. #1036
READING, MA 01867
(617) 644 0014

ENGINEER: CAP DESIGN GROUP

60 LANE PARK
BRIGHTON, MA 02134

FIRE PROTECTION
TIER 1

PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	JK
SCALE	1/4 = 1'-0"
DATE	04/11/25
REVISED	
SHEET	FP